

# ■ FlowMind AI – Capstone Implementation Progress

Weekly Development & Engineering Progress Report • September 2026

|                                                                                                                               |                                                           |
|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>Project Repository:</b><br><a href="https://github.com/Furkhan-5/FlowMind-AI">https://github.com/Furkhan-5/FlowMind-AI</a> | <b>Architecture:</b> Next.js 14, TypeScript, Tailwind CSS |
| <b>Developer:</b> Furkhan (Admin)                                                                                             | <b>Status:</b> Week 1 & Week 2 Fully Completed (100%)     |

## 1. Executive Summary & Vision

FlowMind AI is an enterprise-grade **Multilingual Autonomous AI Operating System** designed to manage end-to-end business workflows across 15 specialized domain agents. It features a modern Pinterest BloomFi-inspired visual aesthetic, multi-script Indic language auto-detection, native speech synthesis, and real-time canonical intent parsing.

## 2. Week 1 Implementation Highlights: UI & Agent Foundation

### Visual Design & Interactive Floating Assistant:

- **Pinterest BloomFi Aesthetics:** Implemented a clean, soft periwinkle background (#eef0f5), 32px rounded white cards, dark navy primary panels (#16132a), and vibrant purple accent tags.
- **Floating Robot Assistant Widget (RobotAssistant.tsx):** Built a bottom-right floating widget featuring a 3D animated Robot launcher button and an interactive 15-agent selection grid.
- **15 Specialized Domain Agents Catalog:** Integrated agents covering CEO, Sales, Marketing, Finance, HR, Support, Analytics, Database, Workflow, Scheduler, Document, Email, Knowledge RAG, Reporting, and Security.
- **SSR-Safe 3D Canvas Rendering:** Handled Three.js WebGL canvas imports using Next.js dynamic loading (ssr: false) to ensure zero server-side hydration errors.

## 3. Week 2 Implementation Highlights: Multilingual Engine & Voice Foundation

- **Indic Script Auto-Detection (indicEngine.ts):** Built Unicode codepoint regex range matching for Telugu (0C00-0C7F), Hindi/Devanagari (0900-097F), Tamil (0B80-0BFF), Kannada (0C80-0CFF), Malayalam (0D00-0D7F), and English.
- **Native Accent Speech Synthesis (TTS):** Connected BCP-47 voice tags (te-IN, hi-IN, ta-IN, kn-IN, ml-IN, en-US) to Web Speech synthesis for regional audio feedback.
- **Canonical Business Intent Parser (intentParser.ts):** Normalized multi-script natural language queries into structured intent actions (SCHEDULE\_SALES\_MEETING, GENERATE\_CLIENT\_INVOICE, APPROVE\_EMPLOYEE\_LEAVE, etc.) with parameters and action confirmation cards.
- **Real-Time Audio Spectrum Visualizer (VoiceSpectrum.tsx):** Developed a 24-bar HTML5 canvas frequency visualizer animated in purple-blue-emerald gradients during speech input/output.
- **Live Browser Web Speech (STT) Microphone Typing:** Integrated continuous Web Speech Recognition streaming spoken text directly into the chat input field live in real time.

## 4. Technical Verification & Build Status

| Verification Check | Executed Command | Status & Result |
|--------------------|------------------|-----------------|
|--------------------|------------------|-----------------|

|                     |                                   |                                                                             |
|---------------------|-----------------------------------|-----------------------------------------------------------------------------|
| TypeScript Compiler | <code>npm run type-check</code>   | <b>PASSED</b> (0 errors)                                                    |
| Next.js Build       | <code>npm run build</code>        | <b>PASSED</b> (Exit Code 0)                                                 |
| Git Push            | <code>git push origin main</code> | <b>COMMITTED</b> (Commit 24841a4)                                           |
| Local Host Server   | <code>npm run dev</code>          | <b>ACTIVE</b> ( <a href="http://localhost:3000">http://localhost:3000</a> ) |

## 5. Roadmap: Upcoming Weeks

### Week 3: Multi-Agent Orchestration & RBAC Security Engine

- Implement CEO Master Orchestrator DAG pipeline execution across Sales, Finance, & Security.
- Enforce Role-Based Access Control (RBAC) role checks (ADMIN, MANAGER, ANALYST, OPERATOR).

### Week 4: Firebase Integration & Cloud Production Deployment

- Connect Firebase Authentication and Firestore DB for chat transcript and action card persistence.
- Deploy cloud backend server endpoints for 100% cross-device speech reliability.